



Area and perimeter

- 1 A rectangle has the dimensions shown below. Find its area and perimeter.

$$\text{Area} = 12 \times 7 = 84 \text{ square units. Perimeter} = 2(12 + 7) = 38 \text{ units.}$$

- 2 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

$$\text{Area} = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66.$$

- 3 A triangle has base 15 and height 10. Find its area.

$$\text{Area} = \frac{1}{2}(15)(10) = 75.$$

The Pythagorean theorem

- 4 A right triangle has the two legs shown below. Find the length of the hypotenuse c .

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

- 5 A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

$$h = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24 \text{ feet.}$$

Volume and surface area

- 6 Find the volume of a cylinder with radius 4 and height 9, in terms of π .

$$V = \pi r^2 h = \pi(16)(9) = 144\pi.$$

- 7 Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

$$SA = 2(lw + lh + wh) = 2(20 + 15 + 12) = 2(47) = 94 \text{ square units.}$$

Similar triangles and proportional reasoning

- 8 Triangle ABC is similar to triangle DEF , with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF .

$$\text{Scale factor} = \frac{DE}{AB} = \frac{12}{8} = 1.5. EF = 10 \times 1.5 = 15.$$

- 9 A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

$$\frac{\text{height}}{18} = \frac{5}{3}, \text{ so height} = \frac{18 \times 5}{3} = 30 \text{ feet.}$$

Angle relationships

- 10 Two parallel lines are cut by a transversal. If one angle measures 72° , find its corresponding angle and its co-interior (same-side interior) angle.

The corresponding angle is equal: 72° . The co-interior angle is supplementary: $180^\circ - 72^\circ = 108^\circ$.

- 11 The three angles of a triangle are in the ratio 2 : 3 : 4. Find each angle.

The parts sum to $2 + 3 + 4 = 9$, and the total is 180° , so each part is 20° . The angles are 40° , 60° , and 80° .

- 12 Two angles are complementary, and one is 34° more than the other. Find both angles.

$x + (x + 34) = 90$, so $2x = 56$, giving $x = 28$. The angles are 28° and 62° .

Coordinate geometry: distance and midpoint

- 13 Find the distance between the points $(3, -2)$ and $(9, 6)$.

$$d = \sqrt{(9 - 3)^2 + (6 - (-2))^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

- 14 Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.

$$\text{Midpoint} = \left(\frac{-5 + 7}{2}, \frac{3 + (-9)}{2} \right) = (1, -3).$$

Circles: area, circumference, and equation

- 15 A circle has the radius shown below. Find its area and circumference, in terms of π .

$$\text{Area} = \pi(8)^2 = 64\pi. \text{ Circumference} = 2\pi(8) = 16\pi.$$

- 16 Write the equation of a circle with center $(3, -4)$ and radius 6.

$$(x - 3)^2 + (y + 4)^2 = 36.$$

- 17 A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π .

$$\text{Sector area} = \frac{60}{360} \times \pi(9)^2 = \frac{1}{6} \times 81\pi = 13.5\pi.$$

Composite and shaded regions

- 18** A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π .

Square area = $12^2 = 144$. Circle area = $\pi(5)^2 = 25\pi$. Remaining area = $144 - 25\pi$.

- 19** A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

Garden area = 300. Fountain area $\approx \pi(3)^2 \approx 28.26$. Remaining area $\approx 300 - 28.26 \approx 272$ square meters.

A well-designed word problem: designing a park

- 20** This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π . (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π .

(a) The semicircle's diameter equals the 20-meter width, so the radius is 10 meters. (b) Rectangle area = $30 \times 20 = 600$. Semicircle area = $\frac{1}{2}\pi(10)^2 = 50\pi$. Total area = $600 + 50\pi$ square meters. (c) The fenced sides are the two 30-meter lengths, the far 20-meter end, and the semicircular arc. The arc length is half the circumference: $\frac{1}{2}(2\pi \times 10) = 10\pi$. Total fence = $30 + 30 + 20 + 10\pi = 80 + 10\pi$ meters.

Multiple choice — easy

- 21** A rectangle has length 23 and width 5. Find its area and perimeter.

a. 115, 56 ✓

b. 56, 115

c. 119, 56

d. 115, 54

Area = $23 \times 5 = 115$ square units. Perimeter = $2(23 + 5) = 56$ units.

22 A rectangle has length 5 and width 23. Find its area and perimeter.

a. 115, 56 ✓

b. 56, 115

c. 119, 56

d. 115, 54

Area = $5 \times 23 = 115$ square units. Perimeter = $2(5 + 23) = 56$ units.

23 A rectangle has length 17 and width 15. Find its area and perimeter.

a. 255, 64 ✓

b. 64, 255

c. 259, 64

d. 255, 62

Area = $17 \times 15 = 255$ square units. Perimeter = $2(17 + 15) = 64$ units.

24 A rectangle has length 30 and width 20. Find its area and perimeter.

a. 100, 600

b. 604, 100

c. 600, 98

d. 600, 100 ✓

Area = $30 \times 20 = 600$ square units. Perimeter = $2(30 + 20) = 100$ units.

25 A rectangle has length 27 and width 6. Find its area and perimeter.

a. 162, 66 ✓

b. 66, 162

c. 166, 66

d. 162, 64

Area = $27 \times 6 = 162$ square units. Perimeter = $2(27 + 6) = 66$ units.

26 A rectangle has length 9 and width 6. Find its area and perimeter.

a. 54, 28

b. 58, 30

c. 54, 30 ✓

d. 30, 54

Area = $9 \times 6 = 54$ square units. Perimeter = $2(9 + 6) = 30$ units.

27 A rectangle has length 4 and width 13. Find its area and perimeter.

a. 52, 34 ✓

b. 34, 52

c. 52, 32

d. 56, 34

Area = $4 \times 13 = 52$ square units. Perimeter = $2(4 + 13) = 34$ units.

28 A rectangle has length 30 and width 18. Find its area and perimeter.

a. 540, 94

b. 96, 540

c. 544, 96

d. 540, 96 ✓

Area = $30 \times 18 = 540$ square units. Perimeter = $2(30 + 18) = 96$ units.

29 A rectangle has length 30 and width 24. Find its area and perimeter.

- a. 720, 106
- b. 720, 108 ✓
- c. 108, 720
- d. 724, 108

Area = $30 \times 24 = 720$ square units. Perimeter = $2(30 + 24) = 108$ units.

30 A rectangle has length 26 and width 16. Find its area and perimeter.

- a. 416, 84 ✓
- b. 416, 82
- c. 420, 84
- d. 84, 416

Area = $26 \times 16 = 416$ square units. Perimeter = $2(26 + 16) = 84$ units.

31 A rectangle has length 23 and width 3. Find its area and perimeter.

- a. 69, 50
- b. 52, 69
- c. 69, 52 ✓
- d. 73, 52

Area = $23 \times 3 = 69$ square units. Perimeter = $2(23 + 3) = 52$ units.

32 A rectangle has length 20 and width 15. Find its area and perimeter.

- a. 304, 70
- b. 70, 300
- c. 300, 68
- d. 300, 70 ✓

Area = $20 \times 15 = 300$ square units. Perimeter = $2(20 + 15) = 70$ units.

33 A rectangle has length 14 and width 25. Find its area and perimeter.

- a. 354, 78
- b. 350, 76
- c. 350, 78 ✓
- d. 78, 350

Area = $14 \times 25 = 350$ square units. Perimeter = $2(14 + 25) = 78$ units.

34 A rectangle has length 13 and width 23. Find its area and perimeter.

- a. 72, 299
- b. 303, 72
- c. 299, 70
- d. 299, 72 ✓

Area = $13 \times 23 = 299$ square units. Perimeter = $2(13 + 23) = 72$ units.

35 A rectangle has length 23 and width 8. Find its area and perimeter.

- a. 184, 60
- b. 62, 184
- c. 184, 62 ✓
- d. 188, 62

Area = $23 \times 8 = 184$ square units. Perimeter = $2(23 + 8) = 62$ units.

Multiple choice — medium

36 A right triangle has legs 3 and 4. Find the length of the hypotenuse c .

a. 5 ✓

b. 7

c. 3

d. 9

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

37 A right triangle has legs 60 and 63. Find the length of the hypotenuse c .

a. 85

b. 89

c. 87 ✓

d. 123

$$c = \sqrt{60^2 + 63^2} = \sqrt{3600 + 3969} = \sqrt{7569} = 87.$$

38 A right triangle has legs 8 and 15. Find the length of the hypotenuse c .

a. 17 ✓

b. 19

c. 23

d. 15

$$c = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

39 A right triangle has legs 7 and 24. Find the length of the hypotenuse c .

a. 27

b. 25 ✓

c. 31

d. 23

$$c = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

40 A right triangle has legs 9 and 40. Find the length of the hypotenuse c .

a. 49

b. 39

c. 41 ✓

d. 43

$$c = \sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41.$$

41 A right triangle has legs 20 and 21. Find the length of the hypotenuse c .

a. 27

b. 31

c. 41

d. 29 ✓

$$c = \sqrt{20^2 + 21^2} = \sqrt{400 + 441} = \sqrt{841} = 29.$$

42 A right triangle has legs 24 and 45. Find the length of the hypotenuse c .

a. 53

b. 51 ✓

c. 69

d. 49

$$c = \sqrt{24^2 + 45^2} = \sqrt{576 + 2025} = \sqrt{2601} = 51.$$

43 A right triangle has legs 11 and 60. Find the length of the hypotenuse c .

- a. 71
- b. 63
- c. 61 ✓
- d. 59

$$c = \sqrt{11^2 + 60^2} = \sqrt{121 + 3600} = \sqrt{3721} = 61.$$

44 A right triangle has legs 24 and 70. Find the length of the hypotenuse c .

- a. 72
- b. 74 ✓
- c. 94
- d. 76

$$c = \sqrt{24^2 + 70^2} = \sqrt{576 + 4900} = \sqrt{5476} = 74.$$

45 A right triangle has legs 15 and 36. Find the length of the hypotenuse c .

- a. 51
- b. 37
- c. 39 ✓
- d. 41

$$c = \sqrt{15^2 + 36^2} = \sqrt{225 + 1296} = \sqrt{1521} = 39.$$

46 A right triangle has legs 18 and 80. Find the length of the hypotenuse c .

- a. 98
- b. 84
- c. 82 ✓
- d. 80

$$c = \sqrt{18^2 + 80^2} = \sqrt{324 + 6400} = \sqrt{6724} = 82.$$

47 A right triangle has legs 40 and 42. Find the length of the hypotenuse c .

a. 58 ✓

b. 56

c. 60

d. 82

$$c = \sqrt{40^2 + 42^2} = \sqrt{1600 + 1764} = \sqrt{3364} = 58.$$

48 A right triangle has legs 27 and 120. Find the length of the hypotenuse c .

a. 125

b. 123 ✓

c. 121

d. 147

$$c = \sqrt{27^2 + 120^2} = \sqrt{729 + 14400} = \sqrt{15129} = 123.$$

49 A right triangle has legs 12 and 35. Find the length of the hypotenuse c .

a. 35

b. 37 ✓

c. 39

d. 47

$$c = \sqrt{12^2 + 35^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37.$$

50 A right triangle has legs 9 and 12. Find the length of the hypotenuse c .

- a. 13
- b. 21
- c. 15 ✓
- d. 17

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

Multiple choice — hard

51 Find the volume of a cylinder with radius 7 and height 13, in terms of π .

- a. 2548π
- b. 643π
- c. 91π
- d. 637π ✓

$$V = \pi r^2 h = \pi(49)(13) = 637\pi.$$

52 Find the volume of a cylinder with radius 6 and height 4, in terms of π .

- a. 576π
- b. 24π
- c. 150π
- d. 144π ✓

$$V = \pi r^2 h = \pi(36)(4) = 144\pi.$$

53 Find the volume of a cylinder with radius 5 and height 13, in terms of π .

- a. 65π
- b. 1300π
- c. 325π ✓
- d. 331π

$$V = \pi r^2 h = \pi(25)(13) = 325\pi.$$

54 Find the volume of a cylinder with radius 3 and height 6, in terms of π .

- a. 216π
- b. 18π
- c. 60π
- d. 54π ✓

$$V = \pi r^2 h = \pi(9)(6) = 54\pi.$$

55 Find the volume of a cylinder with radius 9 and height 7, in terms of π .

- a. 2268π
- b. 63π
- c. 573π
- d. 567π ✓

$$V = \pi r^2 h = \pi(81)(7) = 567\pi.$$

56 Find the distance between the points $(-2, -7)$ and $(-14, -2)$.

- a. 11
- b. 17
- c. 15
- d. 13 ✓

$$d = \sqrt{((-14) - (-2))^2 + ((-2) - (-7))^2} = \sqrt{144 + 25} = \sqrt{169} = 13.$$

57 Find the distance between the points $(-10, 2)$ and $(-15, -10)$.

a. 13 ✓

b. 11

c. 15

d. 17

$$d = \sqrt{((-15) - (-10))^2 + ((-10) - 2)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

58 Find the distance between the points $(5, -8)$ and $(29, -15)$.

a. 23

b. 27

c. 31

d. 25 ✓

$$d = \sqrt{(29 - 5)^2 + ((-15) - (-8))^2} = \sqrt{576 + 49} = \sqrt{625} = 25.$$

59 Find the distance between the points $(-5, -9)$ and $(-14, 31)$.

a. 41 ✓

b. 43

c. 39

d. 49

$$d = \sqrt{((-14) - (-5))^2 + (31 - (-9))^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41.$$

60 Find the distance between the points $(5, 9)$ and $(-9, 57)$.

a. 50 ✓

b. 52

c. 62

d. 48

$$d = \sqrt{((-9) - 5)^2 + (57 - 9)^2} = \sqrt{196 + 2304} = \sqrt{2500} = 50.$$

61 A square with side length 8 has a circle of radius 2 removed from its center. Find the remaining area, in terms of π .

a. 64π

b. $64 - 4\pi$ ✓

c. $68 - 4\pi$

d. $64 - 6\pi$

$$\text{Square area} = 8^2 = 64. \text{ Circle area} = \pi(2)^2 = 4\pi. \text{ Remaining area} = 64 - 4\pi.$$

62 A square with side length 19 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π .

a. $365 - 25\pi$

b. $361 - 25\pi$ ✓

c. $361 - 27\pi$

d. 361π

$$\text{Square area} = 19^2 = 361. \text{ Circle area} = \pi(5)^2 = 25\pi. \text{ Remaining area} = 361 - 25\pi.$$

- 63 A square with side length 12 has a circle of radius 2 removed from its center. Find the remaining area, in terms of π .

- a. 144π
- b. $144 - 4\pi$ ✓
- c. $148 - 4\pi$
- d. $144 - 6\pi$

Square area $= 12^2 = 144$. Circle area $= \pi(2)^2 = 4\pi$. Remaining area $= 144 - 4\pi$.

- 64 A square with side length 16 has a circle of radius 4 removed from its center. Find the remaining area, in terms of π .

- a. $260 - 16\pi$
- b. 256π
- c. $256 - 18\pi$
- d. $256 - 16\pi$ ✓

Square area $= 16^2 = 256$. Circle area $= \pi(4)^2 = 16\pi$. Remaining area $= 256 - 16\pi$.

- 65 A square with side length 8 has a circle of radius 3 removed from its center. Find the remaining area, in terms of π .

- a. $64 - 9\pi$ ✓
- b. $68 - 9\pi$
- c. 64π
- d. $64 - 11\pi$

Square area $= 8^2 = 64$. Circle area $= \pi(3)^2 = 9\pi$. Remaining area $= 64 - 9\pi$.